

Improved Bounds for Górak's Midpoint Constant

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Abstract

For a bijective M -bi-Lipschitz map T between real Banach spaces, let $A(M)$ be the largest possible midpoint defect divided by the distance between its two endpoints. Górak's open constant is

$$K = \limsup_{\varepsilon \downarrow 0} \frac{A(1 + \varepsilon)}{\varepsilon}.$$

We prove that the limit in this definition actually exists and that

$$\frac{e}{2} \leq K \leq e.$$

The lower bound is obtained from an absolute-value shear on a specially renormed copy of $\mathbb{R}^2$. The upper bound follows from a new reflection-orbit inequality for $A(M)$, together with the dyadic estimate in Górak's theorem. The argument does not determine the exact value of K .

1 The midpoint modulus

For $M \geq 1$, define

$$A(M) = \sup \frac{\left\| T\left(\frac{a+b}{2}\right) - \frac{T(a)+T(b)}{2} \right\|}{\|a - b\|}, \quad (1)$$

where the supremum is over all real Banach spaces E, F , all bijections $T : E \rightarrow F$ satisfying

$$M^{-1} \|x - y\| \leq \|T(x) - T(y)\| \leq M \|x - y\| \quad (x, y \in E),$$

and all $a \neq b$ in E . Thus, with the notation of Górak's problem,

$$K_\varepsilon = \frac{A(1 + \varepsilon)}{\varepsilon}. \quad (2)$$

The function A is nondecreasing, and the elementary triangle inequality gives

$$0 \leq A(M) \leq \frac{M}{2}. \quad (3)$$

Górak's Theorem 2.1 gives the following dyadic estimate [1]:

$$A(M) \leq \frac{M^{k-1}}{k} \quad (k \in \{2, 4, 8, \dots\}). \quad (4)$$

We will use two consequences. First, if $0 < h \leq 1/2$, one can choose a dyadic k such that $1/2 \leq hk \leq 1$, and hence

$$A(e^h) \leq \frac{e^{(k-1)h}}{k} \leq \frac{e}{k} \leq 2eh. \quad (5)$$

Second, when k itself is dyadic and $h = 1/k$,

$$\frac{A(e^{1/k})}{1/k} \leq e^{1-1/k}. \quad (6)$$

2 A lower bound from a renormed shear

For $t \in \mathbb{R}$, put

$$S_t(x, y) = (x + ty, y).$$

Fix $h > 0$ and define

$$p_h(x, y) = \max\{eh|x|, |y|\}, \quad \|(x, y)\|_h = \sup_{t \in \mathbb{R}} e^{-h|t|} p_h(S_t(x, y)).$$

Lemma 2.1. *The function $\|\cdot\|_h$ is a norm on $\mathbb{R}^2$. Moreover,*

$$e^{-h|s|} \|v\|_h \leq \|S_s v\|_h \leq e^{h|s|} \|v\|_h \quad (s \in \mathbb{R}, v \in \mathbb{R}^2), \quad (7)$$

and

$$\|(1, 0)\|_h = eh, \quad \|(0, 1)\|_h = 1. \quad (8)$$

Proof. For each fixed t , the map $v \mapsto e^{-h|t|} p_h(S_t v)$ is a norm. Their supremum is finite because

$$e^{-h|t|} eh|x + ty| \leq eh|x| + eh|t|e^{-h|t|}|y| \leq eh|x| + |y|,$$

while $e^{-h|t|}|y| \leq |y|$. The supremum is therefore a norm.

Using $S_t S_s = S_{t+s}$ and $|t| \geq |t+s| - |s|$, we obtain

$$\|S_s v\|_h = \sup_{t \in \mathbb{R}} e^{-h|t|} p_h(S_{t+s} v) \leq e^{h|s|} \|v\|_h.$$

Applying this estimate to $S_{-s} S_s v = v$ gives the lower inequality in (7).

Since $S_t(1, 0) = (1, 0)$, the first identity in (8) is immediate. For the second, writing $u = h|t|$ gives

$$\|(0, 1)\|_h = \sup_{u \geq 0} e^{-u} \max\{eu, 1\} = 1,$$

because both e^{-u} and ue^{-u} are at most 1 for $u \geq 0$. □

Proposition 2.2. *For every $h > 0$,*

$$A(e^h) \geq \frac{eh}{2}.$$

Consequently,

$$\liminf_{\varepsilon \downarrow 0} K_\varepsilon \geq \frac{e}{2}.$$

Proof. On $(\mathbb{R}^2, \|\cdot\|_h)$ define

$$T(x, y) = (x + |y|, y).$$

This is a bijection, with inverse $(u, v) \mapsto (u - |v|, v)$. Given two points, write their difference as (r, s) . There is a number $\theta \in [-1, 1]$ such that

$$T(x, y) - T(x', y') = S_\theta(r, s);$$

indeed, for $s \neq 0$ one may take $\theta = (|y| - |y'|)/(y - y')$, and for $s = 0$ take $\theta = 0$. Lemma 2.1 therefore shows that T is e^h -bi-Lipschitz.

Take $a = (0, 1)$ and $b = (0, -1)$. Their midpoint is 0, while

$$\frac{T(a) + T(b)}{2} = (1, 0).$$

By (8), the midpoint defect is eh and $\|a - b\|_h = 2$. Hence $A(e^h) \geq eh/2$. Since $\varepsilon = e^h - 1$, equation (2) yields

$$K_\varepsilon \geq \frac{eh}{2(e^h - 1)} \longrightarrow \frac{e}{2}.$$

□

3 A reflection-orbit inequality

The next estimate is the main new ingredient.

Proposition 3.1 (Reflection-orbit inequality). *For every $M \geq 1$ and every integer $n \geq 1$,*

$$A(M) \left(2n - 4 \sum_{j=1}^n A(M^{2j}) \right) \leq MA(M^{2n}). \quad (9)$$

Proof. Fix an M -bi-Lipschitz bijection $T : E \rightarrow F$ and a pair of distinct points. Translating the source and target, we may assume that the pair is $x, -x$, that its midpoint is 0, and that $T(0) = 0$. Put

$$u = \frac{T(x) + T(-x)}{2}, \quad r = \|x\|.$$

Let $\Psi : E \rightarrow E$ be reflection about 0, so $\Psi(v) = -v$, and let $\Psi' : F \rightarrow F$ be reflection about u ,

$$\Psi'(w) = 2u - w.$$

Define

$$B = T^{-1}\Psi'T, \quad S = \Psi B.$$

The map B is an involutive M^2 -bi-Lipschitz bijection, and it exchanges x and $-x$. It follows that S is M^2 -bi-Lipschitz and fixes both x and $-x$. Since B and Ψ are involutions,

$$S^{-1} = \Psi S \Psi. \quad (10)$$

For $j \in \mathbb{Z}$, set $x_j = S^j(0)$. Equation (10) gives $x_{-j} = -x_j$. Also, S^j is M^{2j} -bi-Lipschitz and fixes $x, -x$, so its midpoint defect at this pair is exactly $\|x_j\|$. Therefore

$$\|x_j\| \leq 2rA(M^{2j}) \quad (j \geq 1). \quad (11)$$

The identity $B = \Psi S$ gives $B(x_{j-1}) = -x_j$. Since $T(Bv) = 2u - T(v)$, we have

$$T(-x_j) = 2u - T(x_{j-1}).$$

Writing

$$q_j = T(x_j) + T(-x_j),$$

we obtain

$$T(x_j) - T(x_{j-1}) = q_j - 2u.$$

Telescoping from $j = 1$ to n , and using $T(x_0) = T(0) = 0$, gives

$$2nu = \sum_{j=1}^n q_j - T(x_n). \quad (12)$$

Apply the definition of $A(M)$ to T and the pair $x_j, -x_j$. Since $T(0) = 0$,

$$\|q_j\| \leq 4A(M) \|x_j\| \leq 8rA(M)A(M^{2j}), \quad (13)$$

where the second inequality uses (11). Moreover,

$$\|T(x_n)\| \leq M \|x_n\| \leq 2MrA(M^{2n}). \quad (14)$$

Taking norms in (12) and using (13)–(14), we get

$$2n \|u\| \leq 8rA(M) \sum_{j=1}^n A(M^{2j}) + 2MrA(M^{2n}).$$

The midpoint-defect ratio of the original pair is $\|u\|/(2r)$. Choosing T, x with this ratio arbitrarily close to $A(M)$ and then taking the supremum yields

$$2nA(M) \leq 4A(M) \sum_{j=1}^n A(M^{2j}) + MA(M^{2n}),$$

which is equivalent to (9). □

4 Existence of the asymptotic limit and the upper bound

Theorem 4.1. *The limit*

$$L = \lim_{h \downarrow 0} \frac{A(e^h)}{h}$$

exists and satisfies

$$\frac{e}{2} \leq L \leq e.$$

Consequently, the limit $\lim_{\varepsilon \downarrow 0} K_\varepsilon$ exists and

$$\boxed{\frac{e}{2} \leq K \leq e}. \tag{15}$$

Proof. Set

$$f(h) = \frac{A(e^h)}{h}.$$

By (5), there are constants $C > 0$ and $h_0 > 0$ such that

$$A(e^s) \leq Cs \quad (0 < s < h_0). \tag{16}$$

For example, one may take $C = 2e$ and $h_0 = 1/2$.

Let

$$\ell = \liminf_{h \downarrow 0} f(h),$$

and choose a sequence $r_m \downarrow 0$ such that $f(r_m) \rightarrow \ell$. Fix one sufficiently small value $r = r_m$, so that $r < h_0$ and $2Cr < 1$. For $0 < h < r/4$, put

$$n = \left\lfloor \frac{r}{2h} \right\rfloor, \quad \rho = 2nh.$$

Then $r - 2h \leq \rho \leq r$. For $1 \leq j \leq n$, estimate (16) gives

$$\sum_{j=1}^n A(e^{2jh}) \leq C \sum_{j=1}^n 2jh = Chn(n+1).$$

Thus, for all sufficiently small h ,

$$2n - 4 \sum_{j=1}^n A(e^{2jh}) \geq 2n(1 - 2Ch(n+1)) \geq 2n(1 - 2Cr) > 0. \tag{17}$$

Apply Proposition 3.1 with $M = e^h$. Since A is nondecreasing and $\rho \leq r$, equations (17) and (9) give

$$f(h) \leq \frac{e^h r}{\rho(1 - 2Cr)} f(r).$$

Letting $h \downarrow 0$ with r fixed, we obtain

$$\limsup_{h \downarrow 0} f(h) \leq \frac{f(r)}{1 - 2Cr}.$$

Now let $r = r_m \downarrow 0$. The right side tends to ℓ , so

$$\limsup_{h \downarrow 0} f(h) \leq \ell = \liminf_{h \downarrow 0} f(h).$$

Hence $f(h)$ has a limit L .

To bound it from above, take a dyadic integer $k \rightarrow \infty$ and set $h = 1/k$. Equation (6) gives

$$f(1/k) \leq e^{1-1/k} \rightarrow e.$$

Since f has a limit along all $h \downarrow 0$, this proves $L \leq e$. The lower bound $L \geq e/2$ follows from Proposition 2.2.

Finally, writing $1 + \varepsilon = e^h$ and using (2),

$$K_\varepsilon = \frac{A(e^h)}{e^h - 1} = f(h) \frac{h}{e^h - 1} \rightarrow L.$$

Thus the limsup in the original definition is an actual limit and (15) follows. $\square$

5 Status and comparison

Górak recorded the bounds $1/2 \leq K \leq 3$ and asked for the exact value [1]. The supplied partial argument correctly improves these to

$$1 \leq K \leq \frac{e^{\sqrt{2}}}{\sqrt{2}} = 2.9085\dots$$

The present argument further improves the interval to

$$1.3591409\dots = \frac{e}{2} \leq K \leq e = 2.7182818\dots$$

and, independently of the numerical improvement, proves that the defining limsup is a limit.

The remaining factor-two gap is genuine in the present method. The lower example is closely related to the sharp $1/e$ lower bound for the rank-one numerical index of real Banach spaces [2]. This makes $K = e/2$ a natural candidate, but the reflection-orbit argument alone only yields the upper bound e . No claim about the exact value is made here.

References

- [1] R. Górak, *Perturbations of isometries between Banach spaces*, Studia Math. **207** (2011), no. 1, 47–58.
- [2] M. Chica, M. Martín, and J. Merí, *Numerical radius of rank-one operators on Banach spaces*, Q. J. Math. **65** (2014), no. 1, 89–100.